

- Class Test

1. Spatial Filtering for noise removed.

Method: Median filter

Explanation:

A median filter removes salt-and-pepper noise by replacing each pixel with median value of its neighboring pixels.

It preserves image edges better than average filters.

Justification:

- Remove noise effectively
- preserves important image details.

2. Edge Detection:

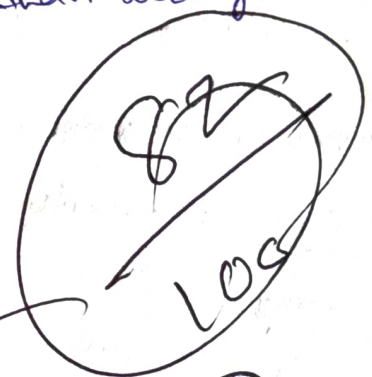
Method: Canny edge detector.

Working:

- Smooth the image using a Gaussian filter
- Calculate gradients.
- Apply double non-maximum suppression
- Use double thresholding and edge tracking.

Effectiveness:

- Detects accurate edges.
- Reduce false edge detection.



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3. Image segmentation:

Method: Region Growing

Explanation:

- Starts from a seed pixel
- Neighboring pixels with similar intensity are added to the region.

Reason:

- Groups adjacent pixels with similar values.
- produces meaningful image regions.

4. Improving Broken Edges:

Method: Morphological closing

Working:

- closing = Dilation followed by Erosion.
- connects broken edges and fills small gaps.

Role:

- Improve edge continuity.
- Enhances object boundaries.

5. Object outline Extractions:

Method: Boundary extraction

Working:

- Perform erosion
- Subtract eroded image from the original image

Result:

- Extract only the object boundary.
- Useful for shape analysis.

6.

Histogram processing:

Method: Histogram equalization

Working:

- Redistributes pixel intensities over the full intensity range
- Improves image contrast.

Output:

- Balanced brightness
- Better visibility of image details.

7.

Image Sharpening

Method: Laplacian filter

Working

- Uses the second derivation to highlight rapid intensity changes.
- Adds the sharpened result to the original image.

Impact:

- Restore fine details
- produces sharper edges.

8. Filtering low-frequency background and high-frequency noise

Method: Bilateral Filter.

Working:

- Smooth homogeneous regions
- preserves edges while reducing noise

Advantages:

- Remove noise
- maintain important structure.

9. Refining Noisy edges.

Method: morphological operations (opening and closing)

Working:

- Opening removes small noisy pixels.
- Closing connects fragmented edges.

Role:

- produces smooth and continuous boundaries.
- Improve segmentation accuracy

10. Precise object boundary extraction.

Method: Canny Edge Detection + watershed segmentation.

Workings:

- Canny accurately detects objects edges
- watershed separated touching objects using detected boundaries.

Effectiveness:

- Accurate objects segmentation.
- precise boundary extraction.